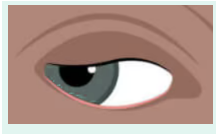
Oculomotor nerve palsy

Oculomotor nerve palsy occurs when the oculomotor nerve (CN III) is compressed or receives inhibited blood flow and is damaged. Because CN III controls voluntary movement in the eye and the parasympathetic action of the pupils, CN III palsy can cause double vision, droopy eyelids, and blown pupils. [1]

Symptoms: [1]



Down+out eye
Drooping eyelid
Blown pupil
Double vision

Causes [1]

Pressure on the nerve → damage to outer fibers

Reduced blood supply to the nerve → damage to inner fibers



The two primary causes of CN III palsy is increased pressure to the nerve and reduced blood supply to the nerve. Increased pressure is typically caused by aneurysms, tumours, or herniation and damages the outer nerve fibers causing symptoms like blown pupils and drooping eyelids. Loss of blood flow typically results from conditions such as diabetes and high blood pressure. It tends to damage the inner nerve fibers which cause symptoms affecting voluntary eye muscle movement such as down and out eye and double vision. [1]

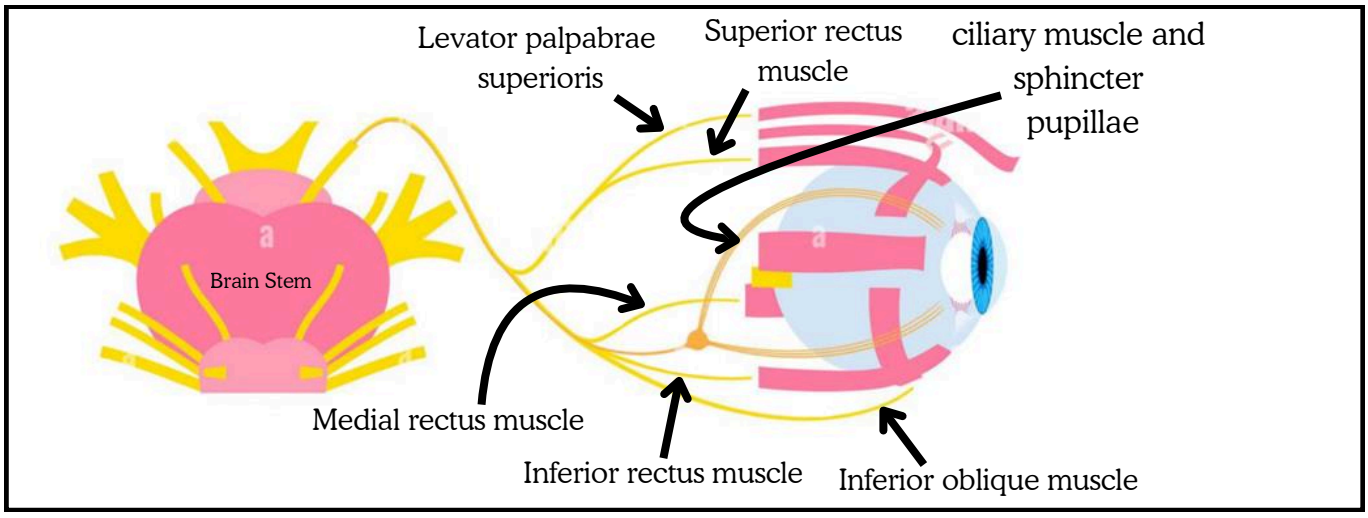


Figure adapted from Alamy.com [2]

Oculomotor Nerve (CN III):

- Responsible for eye movement, eyelid elevation, pupil constriction, and focusing.
- 2 types of motor fibers: voluntary → inner nerve fibers, and involuntary → outer nerve fibers. [3]
- Controls the medial rectus muscle, the inferior rectus muscle, the inferior oblique muscle, and the levator muscle. These muscles control voluntary eye movement.
- Controls the sphincter pupillae and the ciliary muscle, responsible for involuntary movement. [1]

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